

M003 Week 5 — English Worksheet

Topic: Into 3D (x, y, z) · **Course:** 3D Coordinates · **Time:** about 45 minutes

You know the wall (x, y) and the floor (x, z). This week you put them together and use **all three numbers at once: (x, y, z)** — x across, **y up (height)**, z forward. With three numbers you can place a block **anywhere** in space, and build **up**.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
set y to 0
repeat 5 times:
  place stone block at (3, y, 3)
  add 1 to y
```

Which way does this build — across the ground or up into the sky? How tall is it? Which two numbers stay the same?


```
place gold block at (2, 6, 2)
```

y is 6 and there are no blocks below it. Where does the gold block sit — on the ground or in the air?


```
place red block at (0, 0, 0)
place red block at (0, 4, 0)
```

Both blocks share the same x and z. What is different about them? If you stand next to the first block and look up, what do you see?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should build a **tower going up** at $x = 3, z = 3$. Right now it makes a flat line on the ground instead.

```
set n to 0
repeat 4 times:
  place stone block at (3, 0, n)
  add 1 to n
```

Hint: which slot is n in? The height slot is the **middle** one.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should build a tower **up from the ground** at $(2, 0, 2)$. Right now it builds **down into the ground**.

```
set y to 0
repeat 4 times:
  place stone block at (2, y, 2)
  take 1 from y
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should make a line of stone blocks sitting **on the ground**, where $y = 0$. Right now the whole line **floats in the air**.

```
set x to 0
repeat 4 times:
  place stone block at (x, 3, 0)
  add 1 to x
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a **tower** that goes up using a loop, then place a **floating gold block** in the air above it, like a beacon. When you finish, come back here.

Send a photo or video of your tower, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My tower is ... blocks tall because my loop ...

The number that grew was **y**, which means ...

My floating gold block is at (... , ... , ...).

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your tower in the world. Talk through it like you are teaching someone who has only seen flat pictures before. Try to use these words: **x**, **y**, **z**, **height**, **coordinate**.

1. Show your tower from the bottom to the top.
2. Show your code and say which number grows in the loop.
3. Point at the floating gold block and say its full coordinate — all three numbers in order.
4. Say in your own words what each of **x**, **y**, and **z** means.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.